

Trait Adverbs as an Attribute for Genre Classification

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Abstract

Linguistic analysis methods such as Part-Of-Speech (POS) tagging are a popular way to automatically classify documents by genre. These methods are highly accurate, but also resource intensive and slow. In this paper, we investigate the viability of a term frequency-based technique, specifically the counting of trait adverbs to distinguish between genres. We tested this method on two sets of texts written by a historian and a novelist. We conclude that trait adverbs are used at about the same rate in both genres.

1 Introduction

The development of genre classification tools is an important field in NLP. Genres are predefined categories of writing, characterized by a particular style, form or intended use. Genres often have linguistic features which distinguish them from each other. These features can be analyzed by genre classification tools to automatically detect which genre a text belongs to.

In this paper, we will investigate the difference in the rate of trait adverb usage between two writers in the genres of history and fiction. Our aim is to investigate whether trait adverbs are a viable characteristic to use in genre classification.

Trait adverbs modulate the meaning of verbs, giving a subjective assessment of the action described by the verb. Rittman and Wacholder (2008) showed that higher rate of trait adverb use in a text is often correlated with higher perceived subjectivity of said text.

Historians are generally expected to maintain objectivity, presenting historical events in an impartial and factual way, without incorporating their

personal opinions. In contrast, subjective assessments of actions, situations or characters is an integral part of fictional writing. Therefore our hypothesis is that trait adverbs will be used more frequently in the writings of the novelist.

2 Related Work

Ferizis and Bailey (2006) found that genre classification tools that involved POS tagging although extremely accurate, were orders of magnitude slower than other methods, with little accuracy gain. In fact it was shown that POS tagging contributed to 97% of the time spent on classifying a document using these techniques. Instead, a simpler method was used, which approximated the total number of adverbs in a text. The results were nearly as accurate as those obtained using POS tagging.

Qureshi et al. (2019) also investigated the relevance of adverbs in genre classification. Specifically, they found that fiction articles tend to have a higher adverb/adjective ratio. This highlights the potential uses adverbs can have in genre classification tools.

3 Data

In this experiment we counted the amount of trait adverbs in two sets of text, one being the writings of historian Gaston Maspero, and the other of novelist Charles Dickens. The texts were obtained in plaintext on Project Gutenberg. We chose to select the 10 most popular works of each author, based on the amount of downloads from Project Gutenberg, in order to have a relatively representative sample.

In order to obtain the amount of adverbs in the texts, we used a Python script to compare each word in the text with each word in a list of trait adverbs obtained from Rittman and Wa-

cholder (2008), and keep count of each occurrence (Python script is provided on the Github page). However, since we rely on a much simpler method for detecting these adverbs which does not take anything into account other than the presence of the word itself, we removed a few trait adverbs that often occur in uses other than those of a trait adverb (full list of adverbs provided in the Github). The final dependent variable is the total amount of words in each writer's text, divided by the number of occurrences of trait adverbs, giving us the words per adverb.

Pre-processing The raw data from the texts was tokenized, punctuation marks were removed and all letters were converted to lower-case.

Table 1 provides a summary of the data that will be used in this study.

	Words/TAdv
History	#
Fiction	#

Table 1: Overview of the data set. The independent variables are the genres, the independent variable the number of trait adverbs.

4 Results

Table 2 summarizes the findings of this study. Both writers use trait adverbs at approximately the same rate, with history having a slightly higher rate.

	Words/TAdv
History	14706
Fiction	15549

Table 2: Words per trait adverb for each genre.

Discussion Our findings contradict our initial hypothesis that the fiction texts would contain a significantly higher rate of trait adverb use. In fact, the history text shows a slightly higher rate of trait adverb usage. One explanation could be some of the limitations of this study. The amount of trait adverbs in our sample was quite small. Combining this with the fact that the texts of only two specific writers were studied could mean that some writing idiosyncrasies could have skewed the results.

Alternatively, the initial hypothesis of historical texts being inherently more 'objective' than fiction

could be false, or at least not valid for the work of every historian.

5 Conclusion

This study aimed at examining the possibility of using trait adverb counting as a tool for genre discrimination between history and fiction, by testing a hypothesis that historical texts are more 'objective', and thus use less trait adverbs. Based on our findings this does not seem to be the case. It is important for more research to be carried out on this topic, perhaps using larger samples by a variety of writers and a larger set of trait adverbs in order to address some of the limitations of this study.

6 Github

Github Link

7 Reference links and rationale

Ferizis and Bailey (2006):Link

Rationale: I cited this article because it highlights how simple adverb counting can be useful in genre classification.

Rittman and Wacholder (2008):Link

Rationale: I used this article to support my hypothesis that fiction texts will have more trait adverbs.

Qureshi et al. (2019): Link

Rationale: I used this article to show the usefulness of adverbs in genre classification, and to show an interesting alternative approach than the one taken in my project.

References

- Ferizis, G. and Bailey, P. (2006). Towards practical genre classification of web documents. pages 1013–1014.
- Qureshi, M. R., Ranjan, S., Rajkumar, R., and Shah, K. (2019). A simple approach to classify fictional and non-fictional genres. In Ferraro, F., Huang, T.-H. '., Lukin, S. M., and Mitchell, M., editors, *Proceedings of the Second Workshop on Storytelling*, pages 81–89, Florence, Italy. Association for Computational Linguistics.
- Rittman, R. and Wacholder, N. (2008). Adjectives and adverbs as indicators of affective language for automatic genre detection. 1.